

Onondaga County (Syracuse) Sample



Traffic Impact Study

Prepared September 2, 2026 · 43.0500, -76.0800

SIS sample draft — full tier

Traffic Impact Study

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TRAFFIC IMPACT STUDY FOR DOT PROJECTS

Prepared in Accordance With Chapter 5 of the NYSDOT Highway Design Manual (HDM)

Project name	Onondaga County (Syracuse) Sample
PIN	PIN xxxx.xx (to be assigned by NYSDOT)
Project description	Shopping Center / Retail (85 1,000 sqft GLA) within Syracuse MSA.
NYSDOT Region	Region 3 — Syracuse / Central New York
Site coordinates	43.0500°, -76.0800°
Opening year (ETC)	2027

Note:
It is a violation of law for any person, unless they are acting under the direction of a licensed professional engineer, architect, landscape architect, or land surveyor, to alter an item in any way. If an item bearing the stamp of a licensed professional is altered, the altering engineer, architect, landscape architect, or land surveyor shall stamp the document and include the notation "altered by" followed by their signature, the date of such alteration, and a specific description of the alteration.

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

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NYSDOT region: Region 3 — Syracuse / Central New York
Planning group: SMTC (Syracuse Metropolitan Transportation Council)
Regulatory regime: SEQRA (ECL Article 8 / 6 NYCRR Part 617).
SEQRA classification: Not yet determined — lead agency to classify as Type I (6 NYCRR Part 617.4 — likely EIS), Type II (Part 617.5 — exempt), or Unlisted (Part 617.6 — Short EAF + significance finding) before locking the TIS.

1.0 SUMMARY OF TRAFFIC IMPACTS

Capacity:
A capacity analysis was performed per NYSDOT Highway Design Manual Chapter 5. The following capacity improvement measures are cost effective, and are recommended to reduce reoccurring delay: Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s; Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Crash:
A minimum three-year accident history review was attempted; the automated crash ingests returned no records for this site (see Section 4.0). The HDM Chapter 5 §5.3.4 safety screening must be completed manually with Regional Traffic Office data before submittal.

GML §239 referral: no state or county classified roadway was identified within 500 ft of the site via the NYSDOT RDM FeatureServer. The lead agency should still verify against the county's locally-adopted §239 mapping (some counties extend referral to additional facilities — e.g. county-designated scenic byways).

9	2	2	41.0s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

2.0 TRAVEL SPEEDS

Per NYSDOT HDM Chapter 5 §5.2, posted speed limits and actual operating speeds (85th-percentile) along the study network are reported below. Posted-speed-limit values in the third column are auto-ingested from the NYSDOT Roadway Data Management (RDM) FeatureServer (https://gis.dot.ny.gov/hostingny/rest/services/Roadways/RDM_Roadway_Current/FeatureServer) at PDF-generation time — the same source data viewable through the NYSDOT Traffic Data Viewer (<https://gis.dot.ny.gov/html5viewer/?viewer=tdv>). Operating speeds (85th-percentile) are NOT published by NYSDOT and require a project-specific radar / floating-car speed study.

Street	Functional class	Posted speed	85th-pctile operating speed
Near Erie Boulevard East (ERIE BLVD E NY 5) Thompson Road	Urban Principal Arterial Other	40 mph	Speed study required
Thompson Road (THOMPSON RD)	Urban Minor Arterial	30 mph	Speed study required
Erie Boulevard East (Bridge St)	Urban Principal Arterial Other	40 mph	Speed study required
Bridge Street (Bridge St)	Urban Principal Arterial Other	40 mph	Speed study required
Erie Boulevard East	Urban Principal Arterial Other	40 mph	Speed study required
Bridge Street (WIDEWATERS PKWY)	Urban Local	30 mph	Speed study required

3.0 CAPACITY ANALYSIS

Capacity analyses in this report use the SimpleImpactStudies screening solver — a per-approach control-delay model built on the openly-published Webster uniform-delay and Akçelik overflow formulations — reported against the conventional US control-delay bands. Synchro / SimTraffic, HCS, Sidra, or VISSIM may be substituted for formal submittal per NYSDOT Regional Traffic Office preference.

Study network functional-class composition (per NYSDOT RDM): Urban Principal Arterial Other (7); Urban Minor Arterial (1); Urban Local (1). NYSDOT functional class drives access management expectations under HDM Appendix 5A (entrance standards for state highways) and the default K-factor banding under HDM §5.3.

Level of Service thresholds applied below are the conventional US control-delay and density bands for signalized, unsignalized (2-way stop / all-way stop / roundabout), and freeway facilities, as republished in publicly-available state DOT design manuals. Any component $v/c > 1.0$ is LOS F regardless of delay or density.

Signalized intersections — conventional US control-delay bands, sec/veh (as republished in state DOT design manuals); any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤ 10	10-20	20-35	35-55	55-80	> 80

Unsignalized — 2-way stop / all-way stop / roundabout — conventional US control-delay bands, sec/veh; any $v/c > 1.0$ is F.

LOS	A	B	C	D	E	F
sec/veh	≤ 10	10-15	15-25	25-35	35-50	> 50

Freeway basic / weaving / merge-diverge — conventional US density bands, pc/mi/ln; any component $vd/c > 1.00$ is F.

LOS	A	B	C	D	E	F
pc/mi/ln	≤ 11	11-18	18-26	26-35	35-45	> 45

3.1 Growth Rates

Background traffic growth is applied at 1.5% per year over 1 year. This rate is the per-station median CAGR measured from the NYSDOT Traffic Monitoring AADT FeatureServer for Region 3 — Syracuse / Central New York: 0.00%/yr median across 1,492 count stations (rolling actual-count window approximately 2000-2019; 25th-75th percentile range -1.16% to 1.39%). The measured value supersedes the NYSDOT statewide default fallbacks (per HDM Chapter 5: 1.7%/yr Interstate + ramps from the Data Services Bureau, 2.0%/yr all other roads from the Regional Planning Group). The

controlling Regional Planning Group SMTC (Syracuse Metropolitan Transportation Council) may have separately-adopted growth-rate guidance that overrides the measured median for formal submittal — confirm at scoping.

3.2 Existing AADT and DHV

Existing AADT is sourced from the NYSDOT Traffic Data Viewer. Design Hourly Volume (DHV) is reported per HDM Chapter 5 convention as AADT × K-factor; this analysis applies K = 0.10 as the statewide design default — facility-specific K should be substituted from NYSDOT count-station records for formal submittal. The table below grows the existing volume at 1.5%/yr to Estimated Time of Completion (ETC), ETC + 20 years, and ETC + 30 years per HDM Chapter 5 §5.3.

Intersection	2026 AADT	2026 DHV	2027 (ETC) AADT	2027 (ETC) DHV	2047 AADT	2047 DHV	2057 AADT	2057 DHV
Near Erie Boulevard East	18,820	1,882	19,102	1,910	25,728	2,573	29,858	2,986
Thompson Road & Erie Boulevard East	5,480	548	5,562	556	7,491	749	8,694	869
Erie Boulevard East & Bridge Street	18,820	1,882	19,102	1,910	25,728	2,573	29,858	2,986
Bridge Street & Celi Drive	7,080	708	7,186	719	9,679	968	11,233	1,123
Erie Boulevard East & Sutton Place	8,210	821	8,333	833	11,224	1,122	13,025	1,303
Bridge Street & Widewaters Parkway	7,079	708	7,185	719	9,677	968	11,231	1,123
Smith Street & Erie Boulevard East	1,000	100	1,015	101	1,367	137	1,587	159
East Genesee Street & Carlton Drive	999	100	1,014	101	1,366	137	1,585	158
Bridge Street & Enterprise Parkway	7,080	708	7,186	719	9,679	968	11,233	1,123

Source: AADT estimated from engine peak-hour entering-volume sum via K = 0.10. NYSDOT Traffic Data Viewer measured AADT is recommended for submittal. Projection applies the 1.5%/yr background growth rate compounded annually.

3.3 Traffic Control Device Data

Existing traffic control devices at each study-network intersection are summarized below. Signal timing data (cycle length, phase splits, offset) and per-approach detection presence should be confirmed against the controlling NYSDOT Regional Traffic Office or municipal traffic-engineering signal record for formal submittal.

Intersection	Control type	Approaches w/ detection
Near Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
Thompson Road & Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
Erie Boulevard East & Bridge Street	Signal (controlling RTO to confirm)	Field verification required
Bridge Street & Celi Drive	Signal (controlling RTO to confirm)	Field verification required
Erie Boulevard East & Sutton Place	Signal (controlling RTO to confirm)	Field verification required
Bridge Street & Widewaters Parkway	Signal (controlling RTO to confirm)	Field verification required
Smith Street & Erie Boulevard East	Signal (controlling RTO to confirm)	Field verification required
East Genesee Street & Carlton Drive	Signal (controlling RTO to confirm)	Field verification required
Bridge Street & Enterprise Parkway	Signal (controlling RTO to confirm)	Field verification required

3.4 Capacity Analysis for Existing No-Build Condition

Existing No-Build conditions reflect current-year peak-hour volumes (no growth, no project trips). Existing measured volumes are grown to opening year 2027 at 1.5%/yr to establish the No-Build baseline for §3.5 comparison.

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Near Erie Boulevard East	E	68.7	E	73.7	1.06

Intersection	Existing LOS	Existing delay (s)	No-Build LOS	No-Build delay (s)	v/c
Thompson Road & Erie Boulevard East	B	16.7	B	16.8	0.31
Erie Boulevard East & Bridge Street	E	68.7	E	73.7	1.06
Bridge Street & Celi Drive	B	18.0	B	18.1	0.40
Erie Boulevard East & Sutton Place	B	19.0	B	19.1	0.46
Bridge Street & Widewaters Parkway	B	18.0	B	18.1	0.40
Smith Street & Erie Boulevard East	B	14.1	B	14.1	0.06
East Genesee Street & Carlton Drive	B	14.1	B	14.1	0.06
Bridge Street & Enterprise Parkway	B	18.0	B	18.1	0.40

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

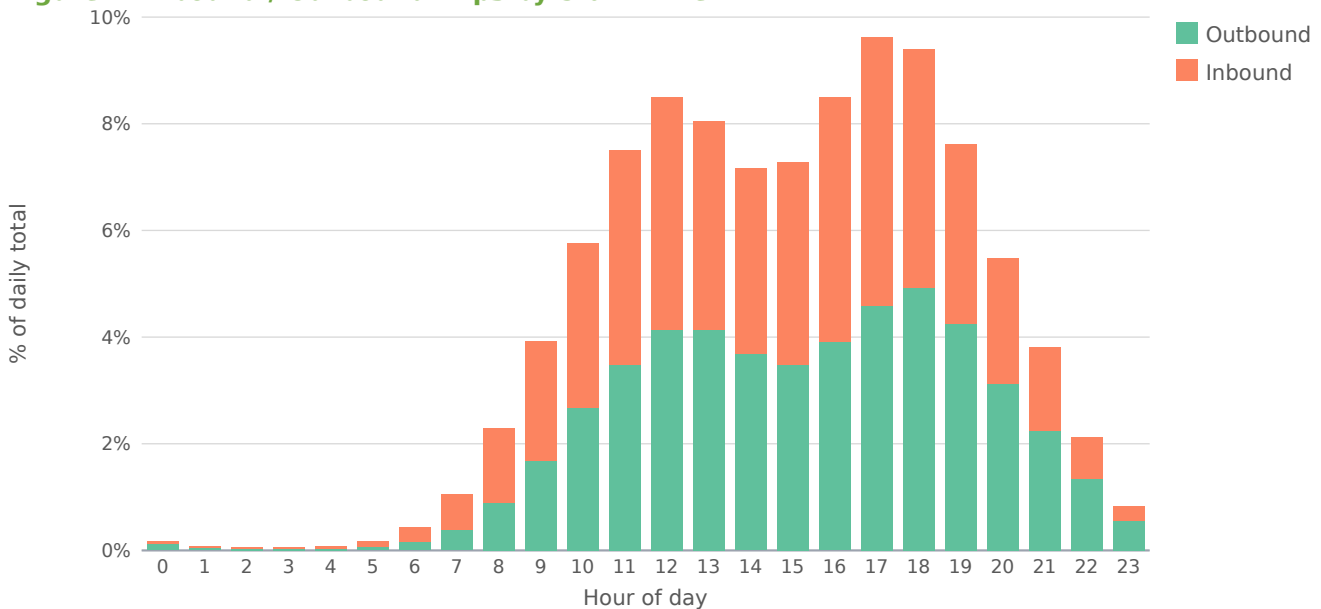
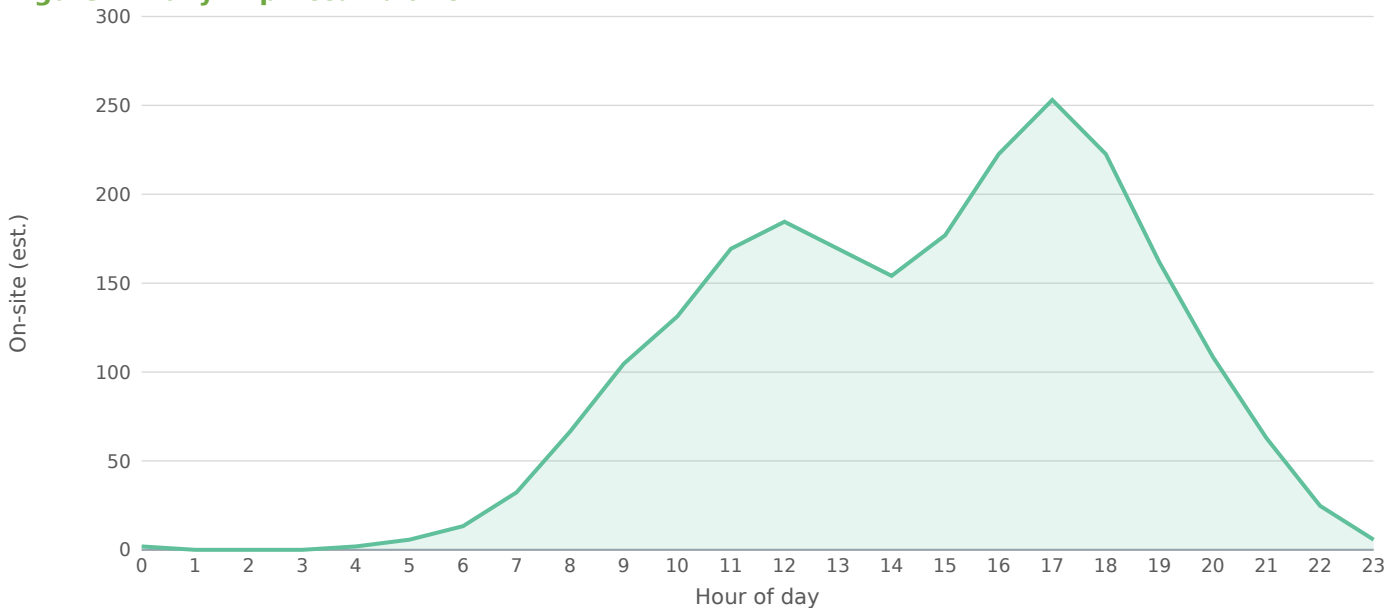


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.5 Capacity Analysis for Proposed Build Condition

Build conditions add the project's external peak-hour trips (680 PM peak, 326 inbound / 354 outbound) to the No-Build baseline at each affected intersection. Per HDM Chapter 5, intersection projects should also analyze each adjacent intersection plus any signal within ½ mile of the project site if expected left-turn volume changes materially affect capacity. The engine's network is determined by a 0.75-mile screening radius; manual scoping should confirm the ½-mile left-turn rule is satisfied.

Intersection	ETC No-Build	ETC Build	ETC+20 NB	ETC+20 Bld	Δ delay (s)
Near Erie Boulevard East		▲ F	F	F	14.6
Thompson Road & Erie Boulevard	B	B	B	B	0.3
East Erie Boulevard East & Bridge Street	E	▲ F	F	F	41.0
Bridge Street & Celi Drive	B	B	C	C	1.6
Erie Boulevard East & Sutton Place	B	B	C	C	0.2
Bridge Street & Widewaters Parkway	B	B	C	C	1.6
Smith Street & Erie Boulevard East	B	B	B	B	0.0
East Genesee Street & Carlton Drive	B	B	B	B	0.0
Bridge Street & Enterprise Parkway	B	B	C	C	0.4

3.6 Capacity Improvement Measures

The following capacity improvements are recommended at affected intersections under Build conditions. Improvement category guidance per HDM Chapter 5: addition of turn lanes; installation or re-timing of traffic signals; signage and striping to prevent unwanted turning movements; conversion to roundabout; and TDM / TSM measures per HDM Chapter 24.

Near Erie Boulevard East [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Erie Boulevard East & Bridge Street [MAJOR]

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

3.7 Trip Distribution — Gravity Model

The trip distribution uses the gravity model ($\text{term} = M / (d^1 \cdot d_{\text{site}})$); mass basis: intersection through-volume (AADT $\times$ K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector		Share of project trips			
Northeast (NNE + ENE)		54%			
Southeast (ESE + SSE)		36%			
Southwest (SSW + WSW)		1%			
Northwest (WNW + NNW)		9%			

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Near Erie Boulevard East	NNE	0.35	1,882	0.3	26.29%
Erie Boulevard East & Bridge Street	ESE	0.42	1,882	0.3	25.64%
Erie Boulevard East & Sutton Place	ESE	0.54	821	0.1	10.78%
Bridge Street & Celi Drive	ENE	0.50	708	0.1	9.39%
Bridge Street & Widewaters Parkway	ENE	0.61	708	0.1	9.07%
Bridge Street & Enterprise Parkway	ENE	0.71	708	0.1	8.79%
Thompson Road & Erie Boulevard East	NNW	0.41	548	0.1	7.52%
Smith Street & Erie Boulevard East	WNW	0.64	100	0.0	1.27%
East Genesee Street & Carlton Drive	SSW	0.71	100	0.0	1.24%

9 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

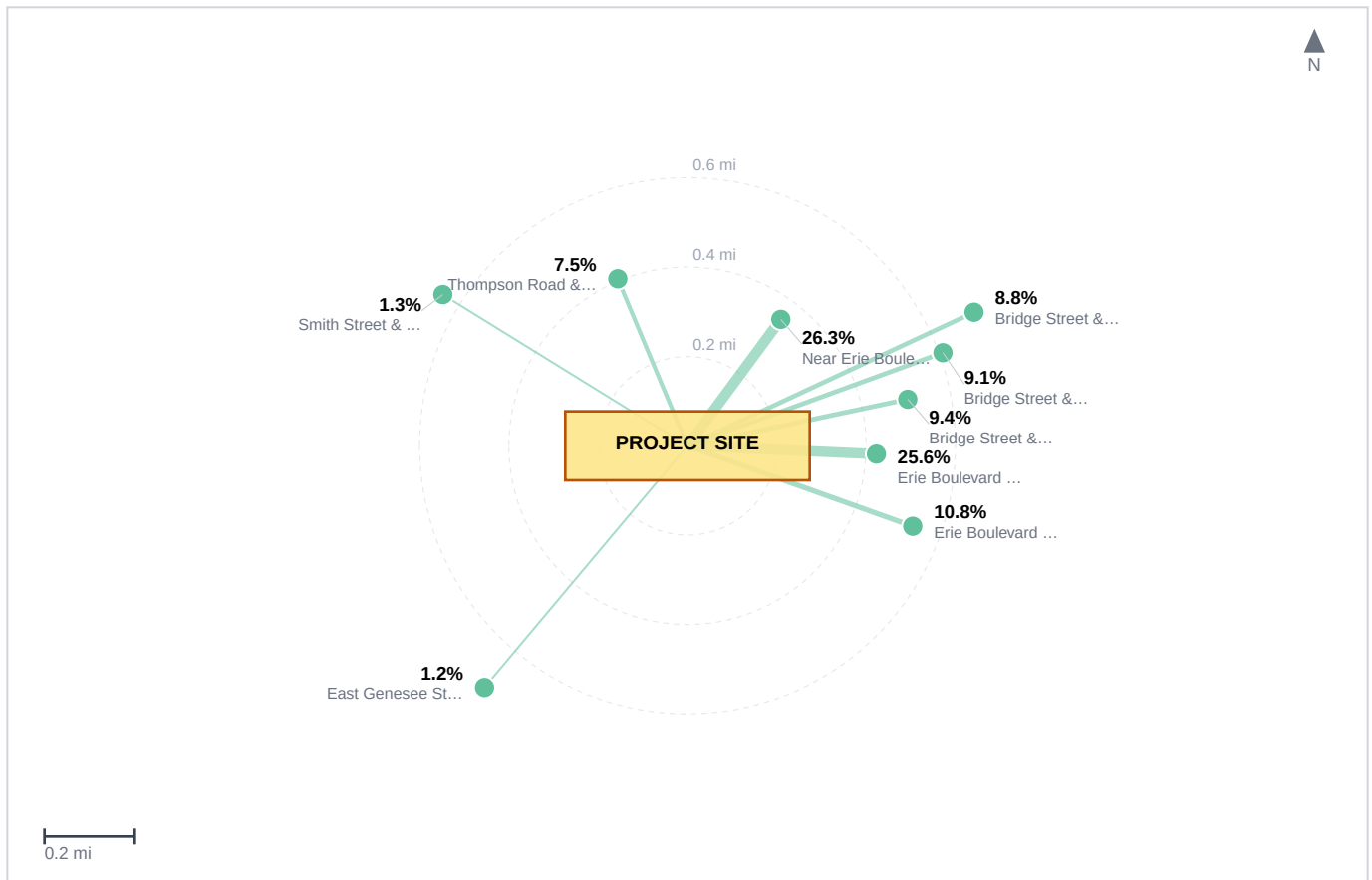
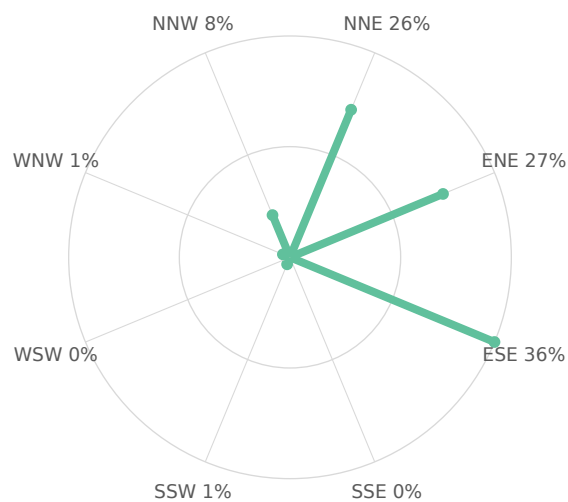


Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone



Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.8 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near Erie Boulevard East	NNE	0.35	40	75	10.1%
Thompson Road & Erie Boulevard East	NNW	0.41	20	37	5.0%
Erie Boulevard East & Bridge Street	ESE	0.42	110	199	26.7%
Bridge Street & Celi Drive	ENE	0.50	100	175	23.5%
Erie Boulevard East & Sutton Place	ESE	0.54	9	24	3.2%
Bridge Street & Widewaters Parkway	ENE	0.61	100	175	23.5%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Smith Street & Erie Boulevard East	WNW	0.64	2	4	0.5%
East Genesee Street & Carlton Drive	SSW	0.71	1	2	0.3%
Bridge Street & Enterprise Parkway	ENE	0.71	37	55	7.4%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

4.0 CRASH ANALYSIS

A minimum three-year accident history review was performed for the period 2023 to 2026 per HDM Chapter 5 §5.3.4. Auto-ingest from the New York State Police 3-year Crash Case Information dataset (data.ny.gov / e8ky-4vqe) was attempted but did not return county-level totals — either the project site's county could not be resolved via the NYSDOT RDM FeatureServer, or the SODA endpoint is unavailable. Refer to Region 3 — Syracuse / Central New York Regional Traffic Office for SIMS output documentation; the Safety Screening criteria from HDM Chapter 5 §5.3.4 (overall accident rate vs. 1.5× statewide average for comparable facility; HAL / PIL / SDL pattern flags; Fatal / Injury / Fatal+Injury rates not above average; addressed PIL list status; Fixed Object, Run-Off Road, and Wet-Road PIL list status) must be applied manually for this project.

APPENDIX A — EXISTING VOLUME REPORT

Per-approach existing peak-hour volumes (vph) for all affected intersections within study limits.
Source: SimpleImpactStudies screening solver (Webster/Akçelik control-delay model), calibrated against the controlling DOT 511 / Regional Traffic Office data feed.

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Near Erie Boulevard East	NB	557	563	604
Near Erie Boulevard East	SB	472	479	496
Near Erie Boulevard East	EB	469	476	476
Near Erie Boulevard East	WB	384	390	409
Thompson Road & Erie Boulevard East	NB	160	162	181
Thompson Road & Erie Boulevard East	SB	141	143	159
Thompson Road & Erie Boulevard East	EB	129	131	134
Thompson Road & Erie Boulevard East	WB	118	120	120
Erie Boulevard East & Bridge Street	NB	637	647	732
Erie Boulevard East & Bridge Street	SB	402	408	521
Erie Boulevard East & Bridge Street	EB	492	499	499
Erie Boulevard East & Bridge Street	WB	352	357	357
Bridge Street & Celi Drive	NB	223	226	288
Bridge Street & Celi Drive	SB	184	186	299
Bridge Street & Celi Drive	EB	164	167	167
Bridge Street & Celi Drive	WB	137	140	140
Erie Boulevard East & Sutton Place	NB	271	275	275
Erie Boulevard East & Sutton Place	SB	212	215	239
Erie Boulevard East & Sutton Place	EB	186	189	189
Erie Boulevard East & Sutton Place	WB	152	155	155
Bridge Street & Widewaters Parkway	NB	190	193	255
Bridge Street & Widewaters Parkway	SB	181	184	296
Bridge Street & Widewaters Parkway	EB	182	185	185
Bridge Street & Widewaters Parkway	WB	155	158	158
Smith Street & Erie Boulevard East	NB	31	31	31
Smith Street & Erie Boulevard East	SB	21	21	22
Smith Street & Erie Boulevard East	EB	27	28	30
Smith Street & Erie Boulevard East	WB	21	22	22
East Genesee Street & Carlton Drive	NB	28	29	29
East Genesee Street & Carlton Drive	SB	25	25	25
East Genesee Street & Carlton Drive	EB	23	24	25
East Genesee Street & Carlton Drive	WB	23	24	25
Bridge Street & Enterprise Parkway	NB	192	195	195
Bridge Street & Enterprise Parkway	SB	185	188	243
Bridge Street & Enterprise Parkway	EB	184	186	186

Intersection	Approach	Existing (vph)	No-Build (vph)	Build (vph)
Bridge Street & Enterprise Parkway	WB	148	150	150

APPENDIX B — EXISTING CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection screening-solver output (Webster/Akçelik control-delay model) for the Existing No-Build condition (current-year volumes; volumes grown to opening year at the \$3.1 background-growth rate for No-Build comparison).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Erie Boulevard East	NB	565	0.70	24.8	C
Near Erie Boulevard East	SB	479	0.59	21.7	C
Near Erie Boulevard East	EB	476	0.59	21.6	C
Near Erie Boulevard East	WB	390	0.48	19.4	B
Thompson Road & Erie Boulevard	NB	162	0.20	15.5	B
East Thompson Road & Erie Boulevard	SB	143	0.18	15.3	B
East Thompson Road & Erie Boulevard	EB	131	0.16	15.1	B
East Thompson Road & Erie Boulevard	WB	120	0.15	15.0	B
East Erie Boulevard East & Bridge Street	NB	647	0.80	29.3	C
Erie Boulevard East & Bridge Street	SB	408	0.50	19.8	B
Erie Boulevard East & Bridge Street	EB	499	0.62	22.3	C
Erie Boulevard East & Bridge Street	WB	357	0.44	18.7	B
Bridge Street & Celi Drive	NB	226	0.28	16.4	B
Bridge Street & Celi Drive	SB	186	0.23	15.8	B
Bridge Street & Celi Drive	EB	167	0.21	15.6	B
Bridge Street & Celi Drive	WB	140	0.17	15.2	B
Erie Boulevard East & Sutton Place	NB	275	0.34	17.2	B
Erie Boulevard East & Sutton Place	SB	215	0.27	16.3	B
Erie Boulevard East & Sutton Place	EB	189	0.23	15.9	B
Erie Boulevard East & Sutton Place	WB	155	0.19	15.4	B
Bridge Street & Widewaters Parkway	NB	193	0.24	15.9	B
Bridge Street & Widewaters Parkway	SB	184	0.23	15.8	B
Bridge Street & Widewaters Parkway	EB	185	0.23	15.8	B
Bridge Street & Widewaters Parkway	WB	158	0.19	15.5	B
Smith Street & Erie Boulevard East	NB	31	0.04	13.9	B
Smith Street & Erie Boulevard East	SB	21	0.03	13.8	B
Smith Street & Erie Boulevard East	EB	28	0.03	13.9	B
Smith Street & Erie Boulevard East	WB	22	0.03	13.8	B
East Genesee Street & Carlton Drive	NB	29	0.04	13.9	B
East Genesee Street & Carlton Drive	SB	25	0.03	13.9	B
East Genesee Street & Carlton Drive	EB	24	0.03	13.9	B
East Genesee Street & Carlton Drive	WB	24	0.03	13.9	B
Bridge Street & Enterprise Parkway	NB	195	0.24	16.0	B
Bridge Street & Enterprise Parkway	SB	188	0.23	15.9	B
Bridge Street & Enterprise Parkway	EB	186	0.23	15.8	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Bridge Street & Enterprise Parkway	WB	150	0.18	15.4	B

APPENDIX C — PROPOSED CONDITION CAPACITY ANALYSIS OUTPUT

Per-intersection screening-solver output (Webster/Akçelik control-delay model) for the Proposed Build condition (No-Build volumes plus project external trips at the assigned distribution).

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Near Erie Boulevard East	NB	604	0.75	26.7	C
Near Erie Boulevard East	SB	496	0.61	22.2	C
Near Erie Boulevard East	EB	476	0.59	21.6	C
Near Erie Boulevard East	WB	409	0.50	19.9	B
Thompson Road & Erie Boulevard	NB	181	0.22	15.8	B
Thompson Road & Erie Boulevard	SB	159	0.20	15.5	B
Thompson Road & Erie Boulevard	EB	134	0.17	15.1	B
Thompson Road & Erie Boulevard	WB	120	0.15	15.0	B
Erie Boulevard East & Bridge Street	NB	732	0.90	38.4	D
Erie Boulevard East & Bridge Street	SB	521	0.64	23.1	C
Erie Boulevard East & Bridge Street	EB	499	0.62	22.3	C
Erie Boulevard East & Bridge Street	WB	357	0.44	18.7	B
Bridge Street & Celi Drive	NB	288	0.36	17.4	B
Bridge Street & Celi Drive	SB	299	0.37	17.6	B
Bridge Street & Celi Drive	EB	167	0.21	15.6	B
Bridge Street & Celi Drive	WB	140	0.17	15.2	B
Erie Boulevard East & Sutton Place	NB	275	0.34	17.2	B
Erie Boulevard East & Sutton Place	SB	239	0.29	16.6	B
Erie Boulevard East & Sutton Place	EB	189	0.23	15.9	B
Erie Boulevard East & Sutton Place	WB	155	0.19	15.4	B
Bridge Street & Widewaters Parkway	NB	255	0.31	16.9	B
Bridge Street & Widewaters Parkway	SB	296	0.37	17.6	B
Bridge Street & Widewaters Parkway	EB	185	0.23	15.8	B
Bridge Street & Widewaters Parkway	WB	158	0.19	15.5	B
Smith Street & Erie Boulevard East	NB	31	0.04	13.9	B
Smith Street & Erie Boulevard East	SB	22	0.03	13.8	B
Smith Street & Erie Boulevard East	EB	30	0.04	13.9	B
Smith Street & Erie Boulevard East	WB	22	0.03	13.8	B
East Genesee Street & Carlton Drive	NB	29	0.04	13.9	B
East Genesee Street & Carlton Drive	SB	25	0.03	13.9	B
East Genesee Street & Carlton Drive	EB	25	0.03	13.9	B
East Genesee Street & Carlton Drive	WB	25	0.03	13.9	B
Bridge Street & Enterprise Parkway	NB	195	0.24	16.0	B
Bridge Street & Enterprise Parkway	SB	243	0.30	16.7	B
Bridge Street & Enterprise Parkway	EB	186	0.23	15.8	B

Intersection	Approach	Vol (vph)	v/c	Delay (s)	LOS
Bridge Street & Enterprise Parkway	WB	150	0.18	15.4	B

APPENDIX D — CRASH ANALYSIS DIAGRAMS AND TABLES

Crash analysis diagrams and tables are not produced by this screening analysis. When the §4.0 crash-analysis section is expanded for formal submittal, the following NYSDOT forms are required: TE-156a (Collision Diagram), TE-164a (Safety Benefits Evaluation Form), TE-204a (Accident Rate Summary), and TE-213 (HAL / PIL / SDL Screening Output). SIMS output and statewide-average rate references should be obtained from the controlling Regional Traffic Office per HDM Chapter 5 §5.3.4 and the NYSDOT Office of Modal Safety statewide rate tables (<https://www.dot.ny.gov/divisions/operating/osss/highway/accident-rates>).

This report is based on the NYSDOT TIS Shell revised on 9/16/2014.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Erie Boulevard East & Bridge Street	0.42	1.0	199	26.7%
Bridge Street & Celi Drive	0.50	1.2	175	23.5%
Bridge Street & Widewaters Parkway	0.61	1.5	175	23.5%
Near Erie Boulevard East	0.35	0.8	75	10.1%
Bridge Street & Enterprise Parkway	0.71	1.7	55	7.4%
Thompson Road & Erie Boulevard East	0.41	1.0	37	5.0%
Erie Boulevard East & Sutton Place	0.54	1.3	24	3.2%
Smith Street & Erie Boulevard East	0.64	1.5	4	0.5%
East Genesee Street & Carlton Drive	0.71	1.7	2	0.3%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 92% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 8% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 100% of study signals lie on a modeled route; highest loaded-link v/c : 0.29. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	1.76	266	0.29

Corridor (road class)	Loaded length mi	PM project vph	v/c
Principal arterial	1.69	228	0.27
Minor arterial	0.39	120	0.21

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 18 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 251 network junctions, max imbalance 0.00 veh. 8 study intersection(s) carry path-derived movements; 1 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

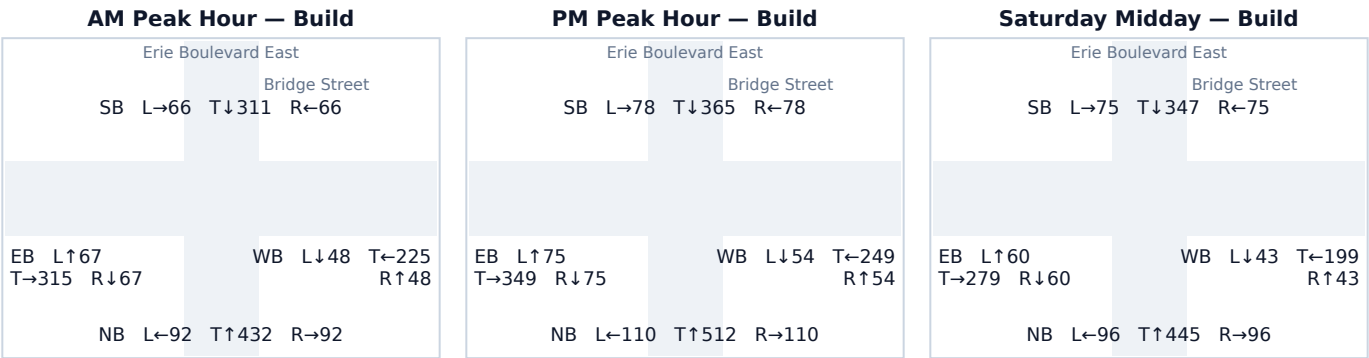
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Erie Boulevard East & Bridge Street

Signal ID	syracuse-592
Location	Central Syracuse · 0.42 mi from site
Added PM peak trips	199
Existing / No-Build (PM)	LOS E · 73.7 s/veh · v/c 1.06
Build (PM)	LOS F · 114.7 s/veh · v/c 1.17
Δ control delay	+41.0 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	647	86	732	0.80 → 0.90	29.3 → 38.4	C → D	700
SB	408	113	521	0.50 → 0.64	19.8 → 23.1	B → C	416
EB	499	0	499	0.62 → 0.62	22.3 → 22.3	C → C	392
WB	357	0	357	0.44 → 0.44	18.7 → 18.7	B → B	252

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	113	57%
NB Through	62	31%
NB Left	24	12%

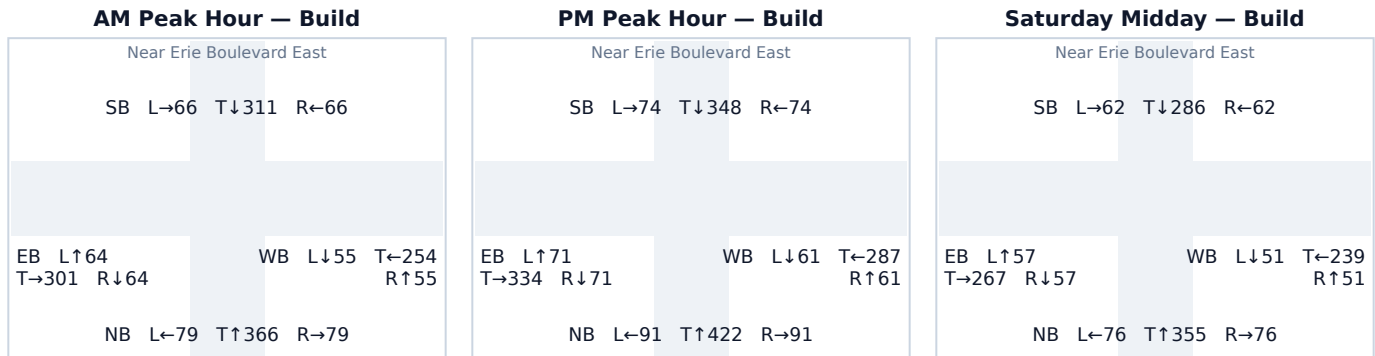
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Near Erie Boulevard East

Signal ID	syracuse-999
Location	Central Syracuse · 0.35 mi from site
Added PM peak trips	75
Existing / No-Build (PM)	LOS E · 73.7 s/veh · v/c 1.06
Build (PM)	LOS F · 88.3 s/veh · v/c 1.10
Δ control delay	+14.6 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	565	39	604	0.70 → 0.75	24.8 → 26.7	C → C	516
SB	479	17	496	0.59 → 0.61	21.7 → 22.2	C → C	389
EB	476	0	476	0.59 → 0.59	21.6 → 21.6	C → C	367
WB	390	19	409	0.48 → 0.50	19.4 → 19.9	B → B	300

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Right	20	27%
WB Left	19	25%
NB Through	19	25%
SB Through	17	23%

Source: geometric octant model (this signal did not resolve to a junction on the modeled road network — no junction within 100 m, or its minor legs are below the network's road-class floor).

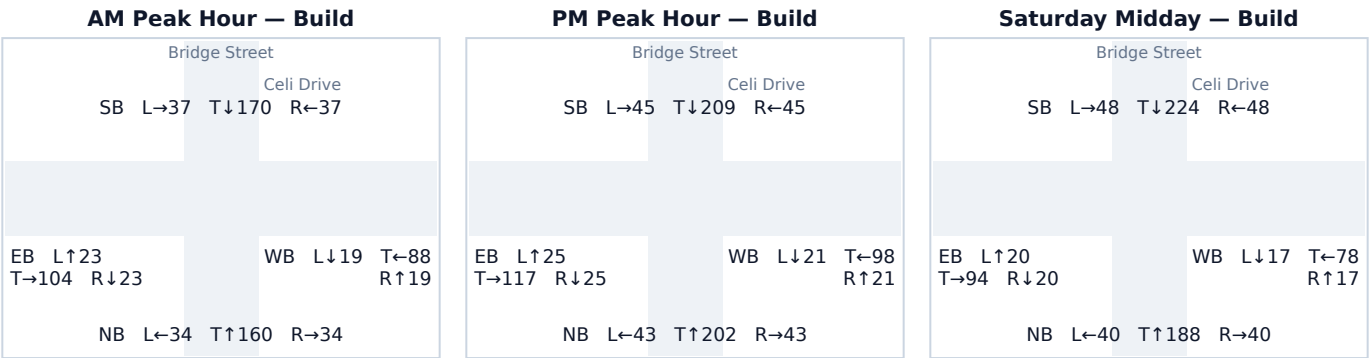
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on

its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Bridge Street & Celi Drive

Signal ID	syracuse-238
Location	Central Syracuse · 0.50 mi from site
Added PM peak trips	175
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 19.7 s/veh · v/c 0.50
Δ control delay	+1.6 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	226	62	288	0.28 → 0.36	16.4 → 17.4	B → B	195
SB	186	113	299	0.23 → 0.37	15.8 → 17.6	B → B	204
EB	167	0	167	0.21 → 0.21	15.6 → 15.6	B → B	104
WB	140	0	140	0.17 → 0.17	15.2 → 15.2	B → B	86

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	113	65%
NB Through	62	35%

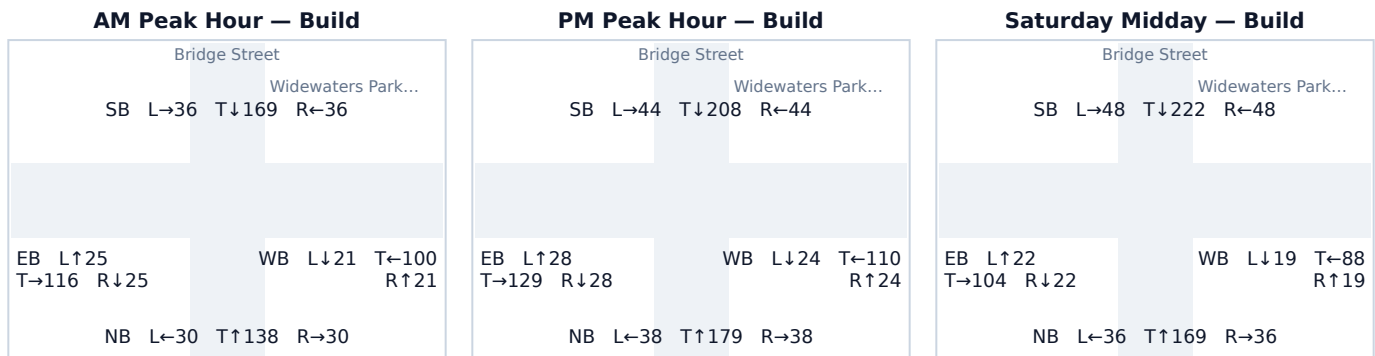
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Bridge Street & Widewaters Parkway

Signal ID	syracuse-924
Location	Central Syracuse · 0.61 mi from site
Added PM peak trips	175
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 19.7 s/veh · v/c 0.50
Δ control delay	+1.6 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count ($AADT \times K\text{-factor}$, $\approx$ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	193	62	255	0.24 → 0.31	15.9 → 16.9	B → B	168
SB	184	113	296	0.23 → 0.37	15.8 → 17.6	B → B	201
EB	185	0	185	0.23 → 0.23	15.8 → 15.8	B → B	117
WB	158	0	158	0.19 → 0.19	15.5 → 15.5	B → B	98

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	113	65%
NB Through	62	35%

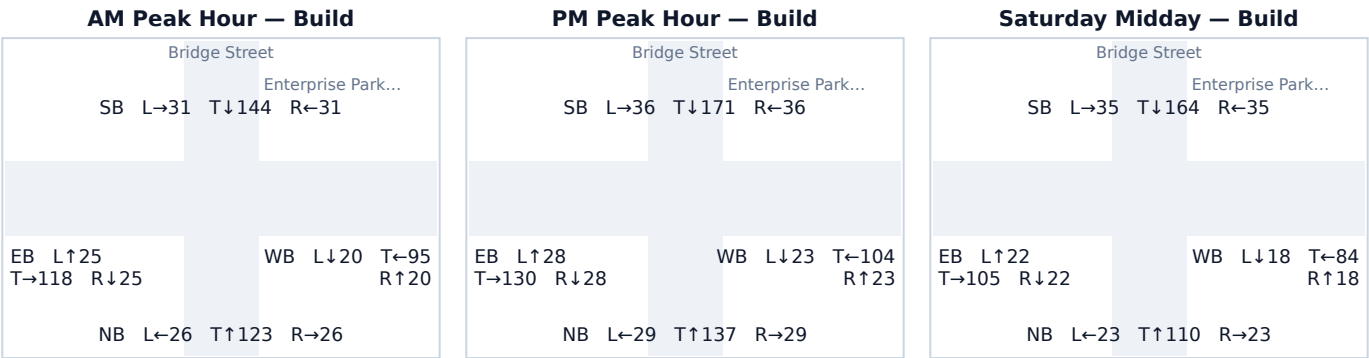
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Bridge Street & Enterprise Parkway

Signal ID	syracuse-920
Location	Central Syracuse · 0.71 mi from site
Added PM peak trips	55
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.5 s/veh · v/c 0.43
Δ control delay	+0.4 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	195	0	195	0.24 → 0.24	16.0 → 16.0	B → B	124
SB	188	55	243	0.23 → 0.30	15.9 → 16.7	B → B	159
EB	186	0	186	0.23 → 0.23	15.8 → 15.8	B → B	118
WB	150	0	150	0.18 → 0.18	15.4 → 15.4	B → B	93

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	55	100%

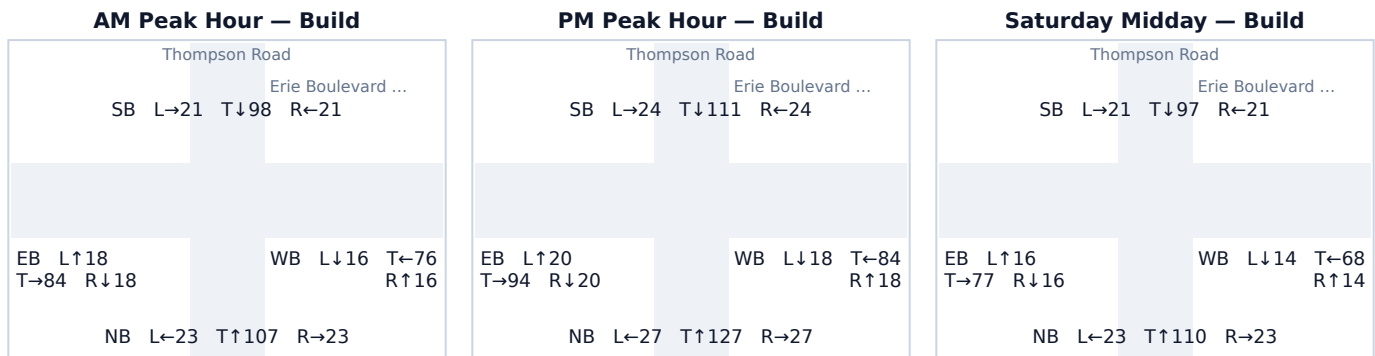
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Thompson Road & Erie Boulevard East

Signal ID	syracuse-774
Location	Central Syracuse · 0.41 mi from site
Added PM peak trips	37
Existing / No-Build (PM)	LOS B · 16.8 s/veh · v/c 0.31
Build (PM)	LOS B · 17.1 s/veh · v/c 0.33
Δ control delay	+0.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	162	18	181	0.20 → 0.22	15.5 → 15.8	B → B	114
SB	143	16	159	0.18 → 0.20	15.3 → 15.5	B → B	99
EB	131	3	134	0.16 → 0.17	15.1 → 15.1	B → B	82
WB	120	0	120	0.15 → 0.15	15.0 → 15.0	B → B	73

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	18	49%
SB Through	16	43%
EB Right	3	8%

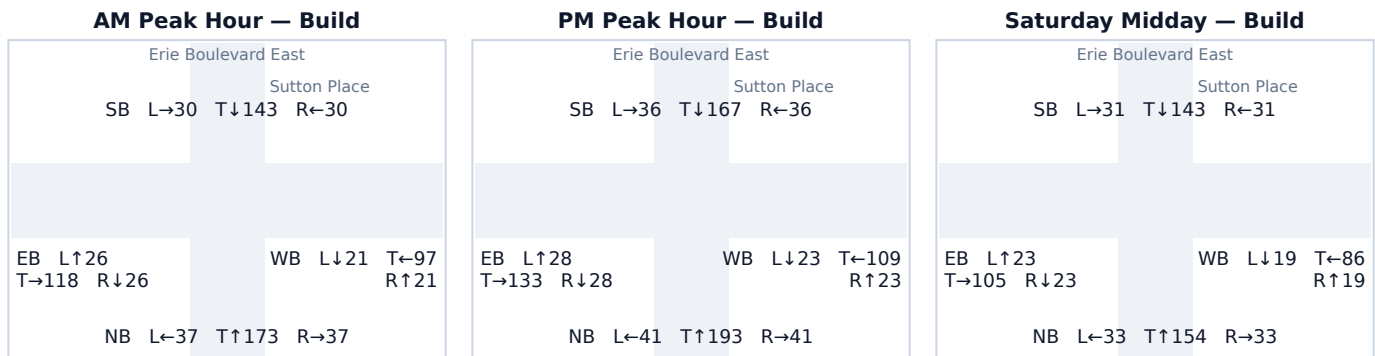
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Erie Boulevard East & Sutton Place

Signal ID	syracuse-588
Location	Central Syracuse · 0.54 mi from site
Added PM peak trips	24
Existing / No-Build (PM)	LOS B · 19.1 s/veh · v/c 0.46
Build (PM)	LOS B · 19.3 s/veh · v/c 0.48
Δ control delay	+0.2 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	275	0	275	0.34 → 0.34	17.2 → 17.2	B → B	184
SB	215	24	239	0.27 → 0.29	16.3 → 16.6	B → B	156
EB	189	0	189	0.23 → 0.23	15.9 → 15.9	B → B	120
WB	155	0	155	0.19 → 0.19	15.4 → 15.4	B → B	96

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	24	100%

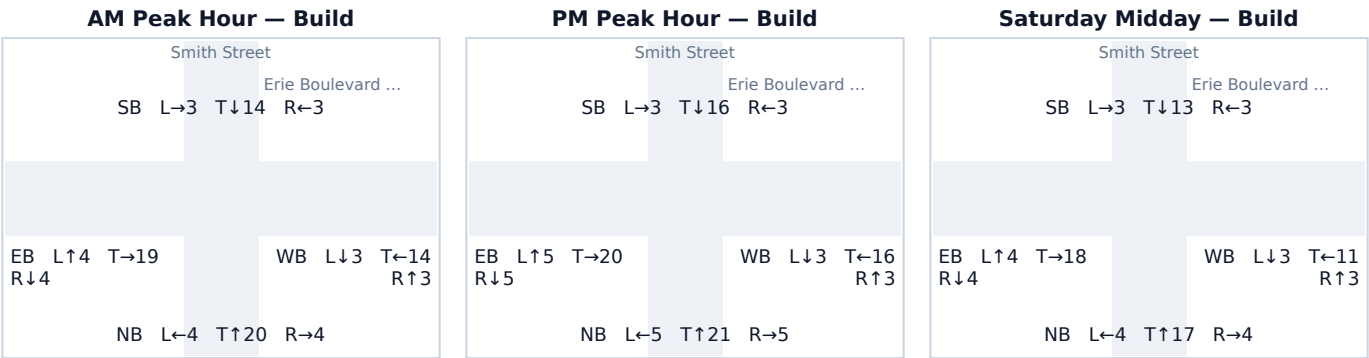
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Smith Street & Erie Boulevard East

Signal ID	syracuse-595
Location	Central Syracuse · 0.64 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	31	0	31	0.04 → 0.04	13.9 → 13.9	B → B	18
SB	21	1	22	0.03 → 0.03	13.8 → 13.8	B → B	13
EB	28	3	30	0.03 → 0.04	13.9 → 13.9	B → B	18
WB	22	0	22	0.03 → 0.03	13.8 → 13.8	B → B	12

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	3	75%
SB Through	1	25%

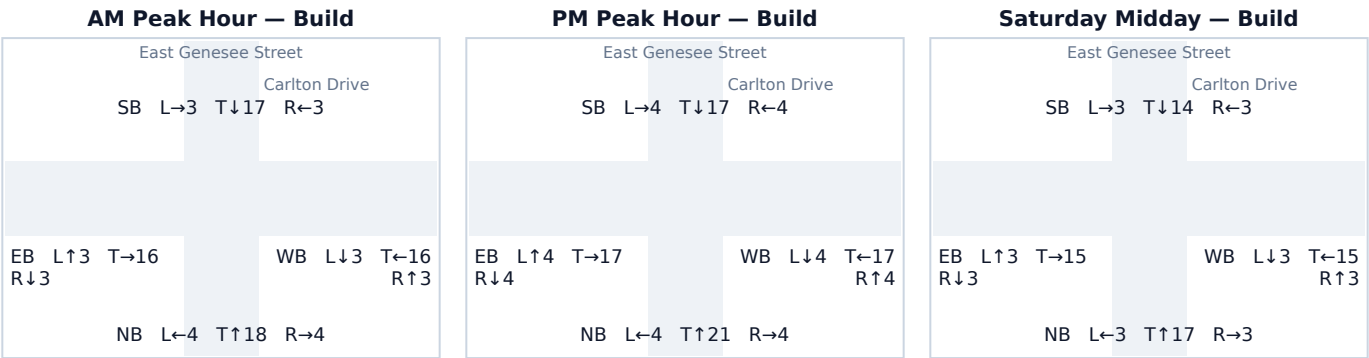
Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 East Genesee Street & Carlton Drive

Signal ID	syracuse-74
Location	Central Syracuse · 0.71 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	29	0	29	0.04 → 0.04	13.9 → 13.9	B → B	17
SB	25	0	25	0.03 → 0.03	13.9 → 13.9	B → B	15
EB	24	1	25	0.03 → 0.03	13.9 → 13.9	B → B	14
WB	24	1	25	0.03 → 0.03	13.9 → 13.9	B → B	14

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	50%
EB Through	1	50%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.